Preliminary Exam 2013
H1 Chemistry Paper 2
Suggested Answers and Comments

1 **(a)** **(i)** PSI of PM10 = $\frac{200-100}{350-150}(320-150)+100$ = 185

PSI of CO = $\frac{300-200}{34-17}(20-17)+100$ = 118 [2]

(ii) Let maximum concentration of CO be x mg/m^3

$\frac{200-100}{17-10}(x-10)+100$ = 150

x = 13.5

$\therefore$ maximum concentration of CO is 13.5 mg m^{-3} [1]

(iii) overall PSI is the maximum value out of 185, 118, 112, 133 and 150.
Hence overall PSI is 185. [1]

(iv) I would advise her not to play for too long. [1]

(b) **(i)** $SO_2 + 2H_2O \rightarrow SO_4^{2-} + 4H^+ + 2e^-$ (x 5)

$2IO_3^- + 12H^+ + 10e^- \rightarrow I_2 + 6H_2O$

overall: $2H^+ + 5SO_2 + 4H_2O + 2IO_3^- \rightarrow 5H_2SO_4 + I_2$

or $5SO_2 + 4H_2O + 2IO_3^- \rightarrow 5SO_4^{2-} + 8H^+ + I_2$ [1]

(ii) $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

n(NaOH) reacted = $0.01 \times 0.005 = 5 \times 10^{-5}$ mol

n(H_2SO_4) reacted with NaOH = $5 \times 10^{-5} \div 2 = 2.5 \times 10^{-5}$ mol
= n(H_2SO_4) produced from SO_2

[1]

n(SO_2) in 1 m^3 sample of air = 2.5×10^{-5}

mass of SO_2 in 1 m^3 sample of air = $2.5 \times 10^{-5} \times 64.0$
= 0.0016 g = 1600 μg

concentration of SO_2 = 1600 μg m^{-3} [1]

(d) **(i)** In 1 m^3,
mass of air is 1 kg
mass of PM10 is $0.00002/100 \times 1 = 2 \times 10^{-7}$ kg = 0.0002 g = 200 μg

Hence concentration of PM10 is 200 μg/m^3 [1]

(ii) Since concentration calculated in **d(i)** is more than 180, the sample of air has exceeded the limit.

The German city of Leipzig will be fined. [1]

2 (a) (i) Mass of CO_2 = 0.15g
No. of moles of CO_2 = 0.15 /44 = 0.003409 mol

Mass of H_2O = 99.85g
Volume of water = 99.85cm^3 [1]

Concentration of CO_2 in 1dm^3 water = 0.003409/ 99.85 x 1000
= 0.0341mol/ dm^3 [1]

(ii) pH = 8.25 => $[H^+] = 10^{-8.25}$

Current $[H^+]$ after 150 years = $0.30(10^{-8.25}) + 10^{-8.25}$ = 7.3×10^{-9} mol dm^{-3}
Current pH = $-\lg (0.30(10^{-8.25}) + 10^{-8.25})$ = 8.14 [1]

(iii) Upon addition of <u>small amount</u> of H^+,
$H^+ + CO_3^{2-} \rightarrow HCO_3^-$ [1]

Upon addition of <u>small amount</u> of OH^-,
$OH^- + HCO_3^- \rightarrow H_2O + CO_3^{2-}$ [1]

(b) (i) Initial $[H^+(aq)] = 10^{-3.68} = 2.09 \times 10^{-4}$ mol dm^{-3} < 0.100 mol dm^{-3} [1]

H_2CO_3 is partially/ weakly dissociated. Hence, H_2CO_3 is a <u>weak acid</u>. [1]

(ii) [KOH] = 25/18.8 x 0.100 = 0.133 mol dm^{-3} [1]

(iii) Phenolphthalein. [1] as the first equivalence point lies within the working range of the indicator of pH 8-10. [1]

3 (a) (i) Element: **W** [1]

(ii) Electronic configuration of Q^+: $1s^2 2s^2 2p^4$
Electronic configuration of Y^+: $1s^2 2s^2 2p^6 3s^2 3p^4$ **[1]**

The election removed from Q^+ is from <u>2p orbital</u>, whereas the electron removed from Y^+ is from <u>3p orbital</u> which is at a <u>higher energy level</u> **[1]**, <u>much lesser energy</u> is required to remove electron from Y^+ than from Q^+ and hence a decrease in second ionisation energy **[1]**.

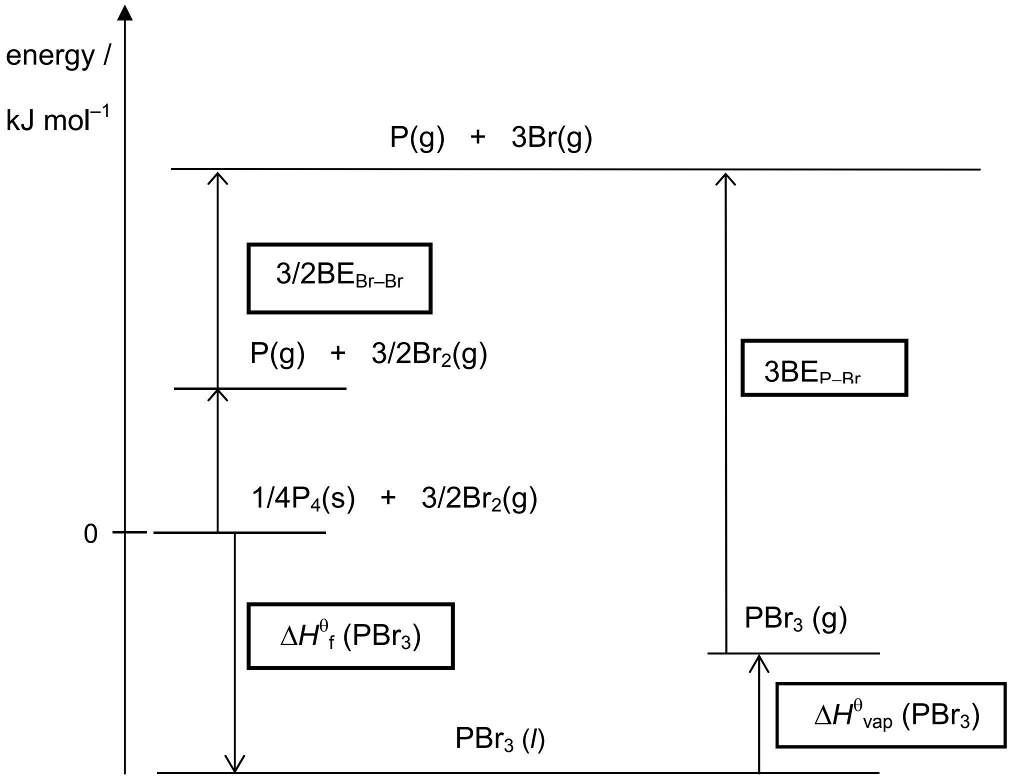
Or

The election removed from Q^+ is from <u>2p orbital</u>, whereas the electron removed from Y^+ is from <u>3p orbital</u>. Since Y^+ has a <u>larger ionic radius</u> than Q^+, the <u>valence e- in Y^+ experience a smaller effective nuclear charge effect and hence less attracted towards nucleus</u> **[1]**. <u>Much lesser energy</u> is required to remove electron from Y^+ than from Q^+ and hence a decrease in second ionisation energy **[1]**.

(iii) VCl_4 [1]

(b) (i) Standard enthalpy change of formation is the amount of heat absorbed or evolved when <u>one mole</u> of a substance is formed <u>from its constituent elements</u>, all in their standard states at <u>298 K and 1 atm</u>. [1]

(ii)



[2] All correct labellings
[1] 2 or 3 correct labellings

Coefficient is not required

(iii) By Hess' Law,

+314.6 + 3/2BE(Br-Br) = -184.5 + (+38.8) + 3BE(P-Br) [1]
+314.6 + 3/2(+193) = -184.5 + (+38.8) + 3BE(P-Br)
3B.E.(P-Br) = + 749.8

Hence B.E. (P-Br) = +250 kJ mol^{-1} [1]

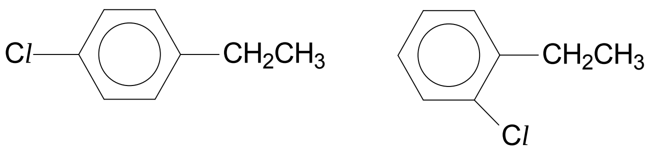
4 (a) (i) oxidation [1]

(ii) nucleophilic [1] substitution [1]

(iii) elimination [1]

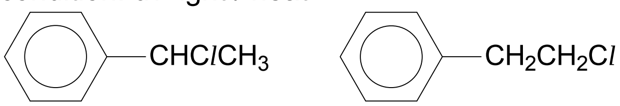
(iv) electrophilic [1] addition [1]

(b) (i) condition: anhydrous $AlCl_3$ / $FeCl_3$ / Fe filings [1]



[1]

(ii) condition: uv light / heat [1]



[1]

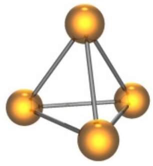
5 (a) (i) First ionisation energy **generally increases** from sodium to argon. [1]
Each electron experiences a **greater effective nuclear charge** due to **increased number of protons** in the nucleus. [1]

(ii) 1st IE of P = 1060 kJ mol^{-1} > 1st IE of S = 1000 kJ mol^{-1}
The 1st IE of sulfur is **lower than expected**. [1]

P: $1s^2 2s^2 2p^6 3s^2 3p^3$ S: $1s^2 2s^2 2p^6 3s^2 3p^4$
The first electron of sulfur to be removed experiences **inter-electronic repulsion**. [1]
Less energy is needed to remove the electron.

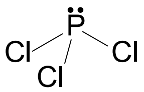
(b) (i) $4\ Ca_5(PO_4)_3F + 18\ SiO_2 + 30\ C \rightarrow 3\ P_4 + 30\ CO + 18\ CaSiO_3 + 2\ CaF_2$ [1]

(ii)



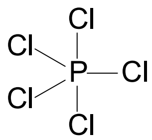
[1]

(c)



[1]

Trigonal pyramidal; 107° [1]



[1]

trigonal bipyramidal;
120° within plane
90° between plane and axis [1]

(d) (i) $K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$ [1] Units: mol dm^{-3} [1]

(ii)

	PCl_5	PCl_3	Cl_2
Initial / mol	$\frac{10}{208.5} = 0.0480$	0	0
Change / mol	-0.45 (0.0480) = -0.0216	+0.45 (0.0480) = +0.0216	+0.45 (0.0480) = +0.0216
Eqm / mol	0.0480 – 0.0216 = 0.0264	0.0216	0.0216
Conc / mol dm^{-3}	$\frac{0.0264}{5} = 5.28\times10^{-3}$	$\frac{0.0216}{5} = 4.32\times10^{-3}$	$\frac{0.0216}{5} = 4.32\times10^{-3}$

[1]
[1]
[1]
[1]

(iii) $K_c = \frac{(4.32 \times 10^{-3})^2}{5.28 \times 10^{-3}} = 3.53 \times 10^{-3}\ \text{mol dm}^{-3}$ [1]

(iv) Total pressure will double. [1]
Position of equilibrium **shifts left to reduce the pressure**. [1]

(Total concentration will increase.
Position of equilibrium shifts left to reduce the total concentration of gases.)

Forward endothermic reaction is favoured. [1]
Position of equilibrium will **shift right to remove the heat energy**. [1]

6 (a) (i) $P_4S_3 + 8O_2 \rightarrow P_4O_{10} + 3SO_2$ [2]

(ignore state symbol; do not accept P_2O_5 and P_2O_3) [ecf for (ii) & (iii)]

[1]: correct oxide
[1]: balanced equation

(ii) Both SO_2 and P_4O_{10} have simple molecular structure with weak van der Waals forces between the molecules [1]
SiO_2 has a giant molecular structure with strong, extensive covalent bonds between Si and O atoms [1]

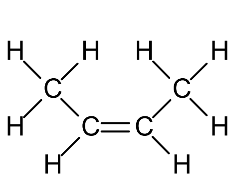
hence SO_2 and P_4O_{10} have m.p. that are similar but much lower than that of SiO_2.

(iii) $SO_2 + H_2O \rightarrow H_2SO_3$ (pH < 2) [2]
$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$ (pH < 2)

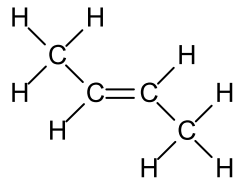
[1] each: balanced equation, include correct pH

(b) (i) **II** : excess concentrated H_2SO_4, 170 / 180 °C [1]

(ii) [3]



cis-but-2-ene

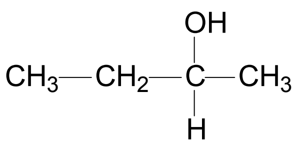


trans-but-2-ene

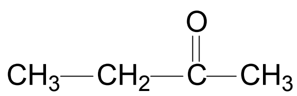
There will be bond strain (or angle strain) for the trans cyclobutene.

[1] each: correctly labelled geometrical isomers
[1]: correct explanation

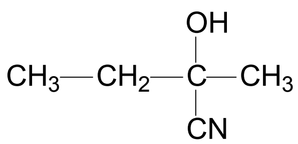
(iii) [9]



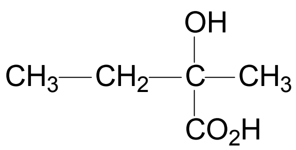
J



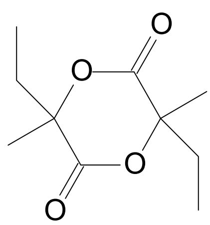
K



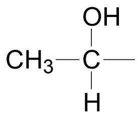
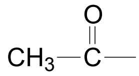
L



M



N

Observation	Type of reaction	Deductions
Compound **J** produce a yellow precipitate on warming with alkaline aqueous iodine.	oxidation (triiodomethane / iodoform test)	**J** contains  OR 
J turned orange hot acidified potassium dichromate green.	oxidation	**J** is an alcohol / (aldehyde) **K** is a ketone / (aldehyde / carboxylic acid)
The product formed is then reacted with HCN, in the presence of NaCN to give compound **L**, C_5H_9NO.	addition	**J** is a 2° alcohol as the product formed from its oxidation (**K**) is a ketone **L** is a cyanohydrin
Treatment of **L** with dilute sulfuric acid gives compound **M**.	hydrolysis	**M** contains a carboxylic acid group
0.590 g of **M** reacts with excess Na to give 120 cm^3 of $H_2(g)$	redox	$n(H_2) = \dfrac{120}{24000} = 5 \times 10^{-3}$ mol $n(\mathbf{M}) = \dfrac{0.590}{118.0} = 5 \times 10^{-3}$ mol Since n(**M**) = $n(H_2)$ evolved, there is presence of 2 –OH group / an alcohol and a carboxylic acid in **M**
On heating **M** in absence of air, compound **N** ($C_{10}H_{16}O_4$) is formed	(self) esterification / condensation	**N** contains (cyclic) ester
N no longer reacts with Na.		Absence of –OH group due to formation of ester in the previous step.

[1] each: structures **J**, **K**, **L**, **M** and **N**
[4]: 10 – 13 pts for explanation
[3]: 7 – 9 pts
[2]: 4 – 6 pts
[1]: 1 – 3 pts

(iv) cyclobutane [1]

7 **(a)** **(i)** The melting generally increases across the period. [1] **[4]**

Na, Mg and A*l* have high melting points because of their giant metallic structures with strong electrostatic forces of attraction between the metal cations and 'sea' of delocalised valence electrons.

Si has an extremely high melting point because it has a giant molecular structure with strong and extensive covalent bonds between the Si atoms.

P_4, S_8, Cl_2 and Ar have simple molecular structures where the discrete molecules are held by weak van der Waals' forces of attraction between them.

[1] each: correct structure and bonding described for each type of structures.

(ii) The number of electrons and hence the strength of Van der Waals' forces of attraction between molecules of S_8 is greater than Cl_2. **[1]**
Thus, more energy is needed to break the stronger Van der Waals' forces of attraction and melting points and boiling points of S_8 is greater than Cl_2.

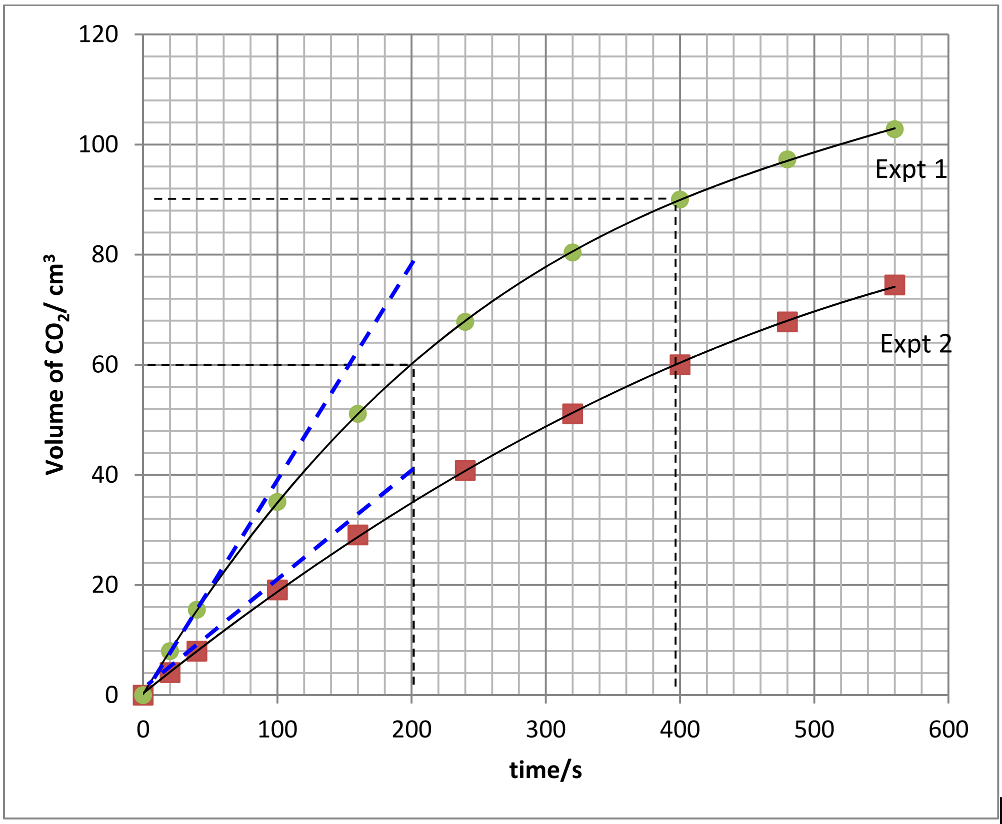
(b) **(i)** Br_2 (aq) + HCOOH (aq) → 2 Br^- (aq) + 2H^+ (aq) + CO_2 (aq)

a = 1, ***b*** = 2, ***c*** = 2 **[1]**

(ii) A catalyst increases the rate of reaction, and hence the rate constant. **[1]**
From the rate equation, rate = *k*[A] where A is the reactant, if rate increases despite all concentrations remaining the same, then the value of *k* must also increase. **[1]**

(iii) Volume = $\frac{20}{1000}$ x 0.25 x 24000 = 120 cm^3 **[1]**

(iv) **[2]**



[1]

correct axes + units
[1] correct shape of curve

(v) *To determine order of reaction w.r.t. Br_2:* **[6]**

Using Experiment **1**,
Half-life is a constant at approximately 200 s.
Thus, reaction is first order with respect to Br_2.

[1] show on graph how $t_{1/2}$ is determined (at least 2 half-lives)
[1] correct conclusion

To determine order of reaction w.r.t. HCOOH:

Initial rate of formation of CO_2 in experiment 1 = $\frac{80-0}{200-0}$ = 0.400 cm^3/s

Initial rate of formation of CO_2 in experiment 2 = $\frac{40-0}{200-0}$ = 0.200 cm^3/s

Let the rate equation be: rate = $k\,[Br_2]^1\,[HCOOH]^x$

$$\frac{0.400}{0.200} = \left(\frac{2.5}{1.25}\right)^x$$

x = 1. Thus, reaction is first order with respect to HCOOH.

Or

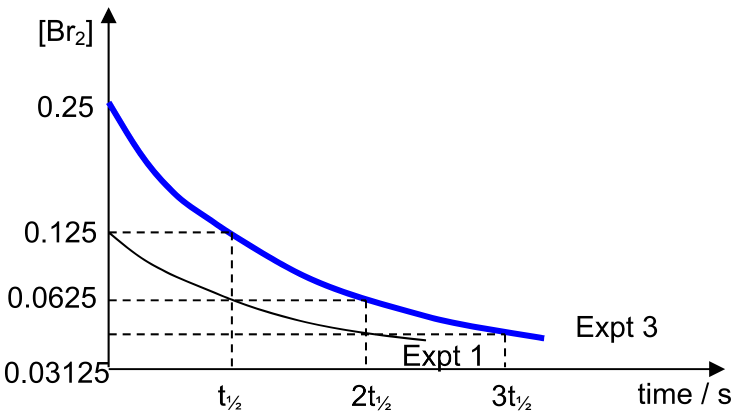
Comparing experiments 1 and 2, when [HCOOH] doubles while keeping [Br_2] constant, the rate of formation of CO_2 doubles.
$\therefore$ reaction is first order with respect to HCOOH.

[1] each: correct calculation of initial rate
[1] each: correct determination of order of reaction (either in words or by calculation, ignore arithmetic error in calculation of initial [HCOOH])
[1] draw tangent to find initial rate

Rate= k [Br_2] [HCOOH]
[1] correct rate equation

(vi) This is so that the concentrations of HCOOH will remain effectively constant throughout each experiment. **[1]**

(vii) **[2]**



[1] Downward sloping graph + correct initial concentration.
[1] $t_{1/2}$ remains constant for both graphs with at least 2 half-lives shown. (shown clearly with corresponding concentration value)